

Self-Healing Cyber-Hardened BMS Presentation Dialogue Script

Companion script for presenting the 3-Slide 7-Layer Presentation Deck

Project: Self-Healing BMS & Digital Twin Cloud Gateway | **GCET 2026**

Slide 1: System Architecture & 7-Layer Blueprint

Estimated Speaking Time: 3 Minutes

"Respected Professors and Members of the Evaluation Committee, welcome to our B.Tech Senior Design Project presentation: 'Self-Healing Cyber-Hardened Battery Management System with Dynamic Trust-Modulated State Estimation, Adaptive Cell Reconfiguration, and Secure Telematics Gateway for Electric Vehicles'. As illustrated on the new architectural blueprint on Slide 1, our project presents a complete 7-Layer Intelligent Engine. The EV Battery Pack represents a modular 16S Li-ion pack of 57.6V nominal voltage (sensing up to 16S via a Texas Instruments BQ76920 battery monitor). The core execution unit is an ESP32 BMS Master featuring dual-core FreeRTOS task separation: Core 0 executes security and AI functions (CAN RX interrupt handler, feature extraction for dt, frequency, variance, and Shannon entropy, and Random Forest ML-IDS computing anomaly scores S from 0.0 to 1.0), while Core 1 handles state estimation and control (I2C AFE sampling, 1RC Extended Kalman Filter with trust-modulated covariance scaling $R_{\text{new}} = R_{\text{base}} * \exp(10 * S)$, SoC/SoH estimation, and cell health monitoring). The 7-Layer Intelligent Engine incorporates: Layer 1 Secure BLE Access (ECDH + AES-128); Layer 2 HMAC Command Authentication (HMAC-SHA256 token authorization); Layer 3 ML CAN Intrusion Detection; Layer 4 Adaptive EKF Trust Modulation; Layer 5 Self-Healing Decision Engine; Layer 6 Cell Reconfiguration & Balancing (controlling passive balancing and cell isolation); and Layer 7 Secure TCU Cloud Gateway streaming telemetry over TLS/MQTT to a real-time Cloud Dashboard. Connected to the 500 kbps CAN bus are an ESP32 Attacker Node (injecting DoS 0x000, Spoofing 0x120, Replay, and Flooding frames) and an ESP32 TCU Node (which reads telemetry, checks anomaly status, formats JSON, and uploads to the cloud dashboard displaying 55.8 V, 12.4 A, Trust Score 0.86, and active self-healing actions)."

Slide 2: Hardware Implementation, BOM & Digital Twin Synchronization

Estimated Speaking Time: 2.5 Minutes

"Moving to Slide 2, we highlight our hardware implementation, Bill of Materials (BOM), and Digital Twin module in visual pictorial form. Our prototype hardware comprises three ESP32 microcontrollers (1,140 INR), one TI BQ76920 AFE IC (520 INR), three SN65HVD230 CAN transceivers (360 INR), a Samsung 18650 battery pack (1,050 INR), passive balancing MOSFETs (75 INR), an OLED display & SD logger (180 INR), and a custom 2-layer KiCad 10 PCB (554 INR). The total prototype cost comes to just 3,879 INR—approximately 46 USD. A key feature highlighted in our new blueprint is the Digital Twin module, which maintains a virtual battery model in software. By receiving real battery data and providing insights, it predicts State of Health (82% SoH prediction), internal resistance growth (+12.4% R_{int}), capacity fade (-8.7%), and predictive thermal rise (43.2 °C predicted for the next 10 minutes). On an ESP-IDF profiler, average CPU utilization is 14.2% on Core 0 and 8.6% on Core 1, with an SRAM memory footprint of 38.4 KB out of 520 KB (a lightweight 7.4% RAM usage)."

Slide 3: How It Works & Self-Healing Active Recovery

Estimated Speaking Time: 3.5 Minutes

"On Slide 3, we detail how our closed-loop self-healing engine operates across four sequential stages and review our verified statistical results and forensic logging capabilities. In Step 1 (Sensor & Twin Sync), the BQ76920 AFE measures cell parameters while the Digital Twin compares live data against virtual expectations. In Step 2 (Core 0 ML-IDS), Core 0 extracts microsecond frame timing, frequency, payload variance, and Shannon entropy, outputting an Anomaly Score S in <0.35 ms. In Step 3 (Layer 4 Adaptive EKF), dynamic covariance scaling $R_{\text{new}} = R_{\text{base}} * \exp(10 * S)$ causes Kalman Gain $K \rightarrow 0$ under attack, isolating corrupt sensor inputs. In Step 4 (Self-Healing & TCU), Layer 5 executes active recovery: continuing normal operation, reducing charging current, disabling cell balancing, entering limp-home mode, or isolating suspect cells, while the TCU ESP32 streams complete forensic security logs (e.g., [10:24:15] DoS Attack Detected -> [10:24:16] Frame Dropped, Trust 0.42 -> [10:24:17] Action: Reduce Charging Current -> [10:24:18] Cell-7 Isolated -> [10:24:20] Uploaded to Cloud). Across 50,000 CAN frames, our evaluation yields: Accuracy of 99.20%, Precision of 98.82%, Recall of 99.51%, F1-Score of 99.16%, Macro ROC-AUC of 0.9942, and PR-AUC of 0.9931. Under active attack, SoC estimation error is bounded within 1.4% (vs 33.84% standard BMS). While our end-to-end FreeRTOS C++ code, ML models, and MATLAB Monte Carlo simulations are 100% verified, physical oscilloscope power profiling and hardware bench trials are marked as 'To Be Tested' for upcoming physical HIL trials. Thank you, and we welcome your questions."
